

Of course. I have processed the new results from the **unclamped** controller.

The difference is night and day. Removing the force clamp allows the controller to use as much power as it deems necessary, leading to a dramatic improvement in performance and noise robustness, but with important caveats.

First, here are the performance metrics organised into tables. Since the averages were not in the output, I have calculated the mean and standard deviation from the three runs for each test.

Unclamped Performance at Noise Level $\sigma = 0$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.02 ± 0.00	0.46 ± 0.00	0.0000 ± 0.0000
-0.9443	0.12 ± 0.00	0.94 ± 0.00	0.0001 ± 0.0000
0.1559	0.01 ± 0.00	0.16 ± 0.00	0.0000 ± 0.0000
-0.6195	0.03 ± 0.00	0.62 ± 0.00	0.0000 ± 0.0000
-0.8419	0.08 ± 0.00	0.84 ± 0.00	0.0001 ± 0.0000

Unclamped Performance at Noise Level $\sigma = 0.001$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.02 ± 0.00	0.46 ± 0.00	0.0002 ± 0.0000
-0.9443	0.12 ± 0.00	0.94 ± 0.00	0.0002 ± 0.0000

0.1559	0.01 ± 0.00	0.16 ± 0.00	0.0002 ± 0.0000
-0.6195	0.03 ± 0.00	0.62 ± 0.00	0.0002 ± 0.0000
-0.8419	0.08 ± 0.00	0.84 ± 0.00	0.0002 ± 0.0000

Unclamped Performance at Noise Level $\sigma = 0.01$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.02 ± 0.00	0.46 ± 0.00	0.0019 ± 0.0002
-0.9443	0.12 ± 0.00	0.94 ± 0.00	0.0019 ± 0.0002
0.1559	0.01 ± 0.00	0.16 ± 0.00	0.0020 ± 0.0002
-0.6195	0.03 ± 0.00	0.62 ± 0.00	0.0017 ± 0.0002
-0.8419	0.08 ± 0.00	0.84 ± 0.00	0.0015 ± 0.0003

Unclamped Performance at Noise Level $\sigma = 0.1$

Initial Angle (rad)	Avg. Stabilisation Time (s)	Avg. Overshoot (rad)	Avg. Steady-State Error (rad)
0.4553	0.02 ± 0.00	0.46 ± 0.00	0.0178 ± 0.0012
-0.9443	0.58 ± 0.01	0.94 ± 0.00	0.0178 ± 0.0003
0.1559	0.01 ± 0.00	0.16 ± 0.00	0.0190 ± 0.0015

-0.6195	0.04 ± 0.01	0.62 ± 0.00	0.0176 ± 0.0020
-0.8419	0.46 ± 0.25	0.84 ± 0.00	0.0168 ± 0.0013

Analysis of Results: Unclamped Controller

1. The Power of Unrestricted Control ($\sigma = 0$)

In the ideal, noise-free scenario, the unclamped controller is phenomenally effective.

- **Blistering Speed:** Stabilisation times are **5 to 10 times faster** than the clamped controller. For the angle at -0.9443 rad, the time dropped from 1.33 seconds to a mere 0.12 seconds. This is because the controller can apply massive initial force to counteract the error instantly, without being limited to 10N.
- **Perfect Precision:** The steady-state error is virtually zero across the board, demonstrating flawless performance when the model is perfect.

2. Astonishing Noise Immunity ($\sigma = 0.01$)

This is the most critical finding. **The unclamped controller succeeds where the clamped controller failed catastrophically.**

- **Stability is Maintained:** At a noise level of $\sigma = 0.01$, where the clamped controller became completely unstable, this controller stabilises just as quickly as it did with no noise. It can generate corrective forces large enough to completely **overpower** the disturbances caused by the noise.
- **Precision Degrades Gracefully:** The steady-state error increases by an order of magnitude (from ~ 0.0002 to ~ 0.0019 rad), which is expected. However, this is a graceful degradation of precision, not a complete failure of stability.

3. Performance Under High Noise ($\sigma = 0.1$)

Even at a very high noise level, the controller remains stable, though the strain begins to show.

- **Variability Emerges:** For the first time, we see significant standard deviation in the stabilisation time (e.g., 0.46 ± 0.25 s for the -0.8419 rad angle). The controller is still winning against the noise, but the fight is much less consistent.
- **Steady-State "Jitter":** The steady-state error increases by another order of magnitude to ~ 0.018 rad. At this stage, the controller's high gains are causing it to aggressively react to every noise spike, leading to significant "jitter" or oscillation around the zero point. It is stable, but not precise.

Clamped vs. Unclamped: A Tale of Two Designs

This comparison reveals a fundamental engineering trade-off between performance and practicality.

- **Clamped Controller (Safe but Brittle):**
 - **Pro:** Operates within safe, realistic physical limits (actuator saturation). It is energy-efficient and won't destroy the hardware.
 - **Con:** Very slow in comparison and extremely sensitive to noise. The limited authority means it cannot overcome disturbances, leading to catastrophic failure.
- **Unclamped Controller (Aggressive but Unrealistic):**
 - **Pro:** Extremely fast and powerful. It is incredibly robust to noise because it can generate whatever force is needed to correct errors.
 - **Con:** This is a purely theoretical advantage. In the real world, this controller would demand infinite power, burn out any motor, and likely break the physical system with its violent, instantaneous force requests.

Conclusion

Removing the force clamp transforms the controller from a fragile, slow system into a brutally effective and robust one **within the simulation**. It highlights that the previous failure was not due to the PID tuning itself, but due to the controller's **authority being insufficient** to reject noise.

However, the unclamped solution is not practical. The real engineering task is to find a middle ground: a controller that **operates within physical limits (clamped)** but is also **robust to**

noise. This is typically achieved not by granting infinite power, but by adding intelligence, such as **filtering the sensor signal before it reaches the PID controller**, to prevent the controller from reacting to noise in the first place.